



Passive Image Forgery Detection Techniques: A Review, Challenges, and Future Directions

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Abstract

The rapid advancement of science and technology has led to the potential for easy manipulation of multimedia content through the use of diverse editing tools. This poses a significant threat to the credibility and integrity of multimedia information. Consequently, substantiating digital images is becoming gradually crucial as digital images hold vital information and are used as essential pieces of evidence in various sectors. The necessity and relevance of digital image forensics have drawn several academics to develop various detection procedures in image forensics. Passive image forgery detection is the foundation of image forensics. Some common passive forgeries that influence the image's authenticity are image splicing, copy-move, and retouching. In recent times, substantial research effort has been devoted to developing novel approaches for detecting several image forgeries. This study provides an overview of similar research efforts that have been carried out utilizing a well-defined methodology. Our goal is to create an efficient way for image forensics researchers to discover new features of forgeries. This study presents a brief introduction to image forensics, including a historical perspective, taxonomy, and framework of image forgery detection approaches. Various resources useful to academic researchers, such as journals, datasets, websites, and performance parameters are explored and presented. This paper will provide a comprehensive review that will aid researchers in overcoming the numerous challenges experienced in earlier studies. Also, future directions are provided to help scholars in this domain. The purpose of this research is to evaluate passive image forgery detection approaches, therefore benefiting new researchers.

Keywords Digital image · Copy-move · Passive image forgery detection · Image splicing

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1 Introduction

Nowadays, there is hardly any platform that does not make use of digital photographs. They are utilized in nearly every sector, including digital media, law, forensics, military, research, social media, glamour, and so on, as well as throughout the internet. With such a large quantity of photographs, the significance of their validity has grown dramatically. Humans are more likely to believe what they see than what they hear. As a result, visual material takes precedence over spoken content in their minds. As a result, people place a high value on what they see in newspapers, magazines, news channels, social media, and other media regularly. Because of their ubiquitous usage, digital photographs are most frequently altered through digital media, distorting their significance with malicious intent. As a result, verifying digital images is critical, making forensic science important [1, 2].

Forensics is the application of knowledge and technology in courtrooms to investigate and verify legitimacy in criminal cases. It is classified into two parts: analog and digital. Figure 1 depicts the forensics taxonomy. Analog forensics emphasizes identifying tangible evidence in reality, which cannot be incorrect or recuse itself. Today, digital forensics that is categorized into multimedia forensics and computer forensics, is the primary focus of forensic research. Computer forensics is applied when computers are employed in

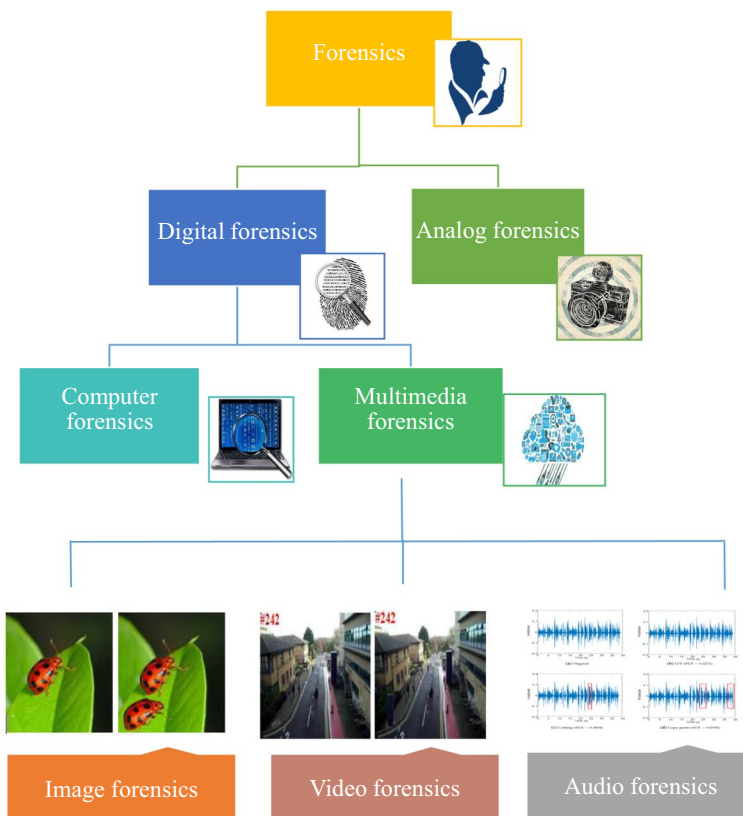


Fig. 1 Taxonomy of forensics

real-world illegal activities and experts seek to retrieve proof from computers. Therefore, this proof should be reliable enough to stand up in court. In terms of digital evidence, multimedia forensics is just like computer forensics; however, in this forensics, digital evidence is linked to the external world and cannot be manipulated using machines. Its goal is to recover some of the misplaced consistency of digital proof by developing tools that provide apparent signs of any modification that happened [3–7].

Image forensics is concerned with identifying proof of counterfeits and primarily with increasing the trustworthiness of digital pictures. It tackles the topic of image authentication or image ancestries and gives reliable responses regarding the foundation and validity of digital photographs. Video forensics detects video tampering, which is subsequently exploited to generate false videos for unlawful reasons. Counterfeiting digital video is harmful since the video has long been a vital tool in struggle against crime. The domain of forensics that deals with the collection, scrutiny, and assessment of audio is known as audio forensics. They can finally be recognized as evidence in criminal law courts or as a source of government inquiries into corruption, accidents, or other civil actions [6–15].

The process of creating false picture by modifying the data of main picture and passing it as original for malicious purposes is known as image forgery. It is an important issue since the importance of digital picture validation has lately acquired a lot of attention since digital multimedia is nowadays widely employed in many safety applications or enterprises. As a result, effective movements for identifying transformed images are being studied.

This research is significant since such discovery algorithms may be employed in high-security businesses, like immigration, where digital pictures and e-passports interrelate, then photographs saved on passport chips must be verified [1, 6, 16].

1.1 Historical Perspective

The practice of image falsification dates back to the nineteenth century. Photographs were being altered within a few decades after Niepce made the first image in 1814. A brief history of image manipulation dating back to the 1800 s is provided. Hippolyte Bayard, who produced a false image in 1840, is famous for a photograph of himself committing suicide. Among the well-known historical counterfeit pictures are: the photograph of Mathew Brady (1865); Satlin removed his adversary from genuine picture (1930), a fake image of Iranian missile (2008), and celebration of Eid al-Adha (2016). Figure 2 depicts an illustration of picture tampering from 1930, with a fake picture on left, the original picture in middle, and an illustration of removed portion on the right. Some of these counterfeit



Fig. 2 Illustration of tampering in 1930, in which Stalin frequently airbrushed his rivals out of pictures. After falling out of favor with Stalin, a commissar was removed from the authentic image

photographs resulted in serious financial and reputation losses, false propaganda, political controversies, extortion, and other undesirable societal consequences [7, 16].

1.2 Problem Statement

- Despite the existence of several suggested approaches in the field of digital forensics to identify forged images, a significant knowledge deficit persists regarding IFDT. Current research mainly focuses on active techniques, leaving a lack of comprehensive evaluations for passive IFDT methods like image splicing, retouching, copy-move forgery, fusion-based, and deep learning-based procedures.
- Prior investigations have either lacked adequate specificity or failed to provide clear directions for following searches. Furthermore, many scholarly articles focus on individual aspects, resulting in a fragmented environment.
- The motive of this article is to address these shortcomings through a comprehensive and unified examination of passive IFDT methods. This will be achieved by integrating information resources, datasets, and performance parameters. This paper analyzes past research flaws to offer clear future research directions in digital image forensics.

1.3 Motivation for Research

The researchers have proposed several strategies for identifying forgery images to re-establish trust in digital images because of several vulnerable problems that are associated with detecting forgery images. Manipulation of photographs has recently become a difficult task as a result of growing tools in digital image forensics, which has boosted belief in digital images. Based on a thorough examination of the existing works, few studies dealing with passive digital IFDT were found. Article presented by Zhang et al. [1] focuses on picture forensics tools for copy-move forgery detection (CMFD). Soni et al. [17] and Warif et al. [18] assessed CMFD advancements and explained CMFD procedure architecture, comprising feature extraction and categorization. Teerakanok et al. [19] shared important insights and current advances in CMFD. Birajdar et al. [20] presented a classification of numerous detection algorithms against CMF and image splicing, as well as detection methods for other image processing operations. Walia et al. [21] and Jindal et al. [2] provided an outline and available references on passive IFDT. Several other surveys [6, 22–27] are also available, but they do not give a comprehensive insight into this domain or a path for future directions. Despite examining the merits and demerits of various detection techniques, such a study fell short of providing a thorough assessment of the present research. Moreover, the majority of published research articles are focused on either CMFD or image splicing. Consequently, there is a need to present numerous passive IFDT comprising of image splicing, retouching, CMF, fusion-based and deep learning-based procedures under one umbrella.

The study in this article varies from the previously stated existing surveys since an efficient strategy is used to perform a comprehensive survey and to offer a thorough description of literature concentrating on various IFDT. In this study, the numerous passive IFDT such as ISF, retouching, CMF, fusion-based and deep learning-based are discussed under one roof. Furthermore, analysis is performed using a variety of factors, including the available datasets, information sources, trends of publications, and performance evaluation criteria. Finally, research issues associated with existing techniques are explored, which

would be valuable for scholars conducting further image forensics research. This study's key contributions can be summed up as follows:

- A thorough evaluation of approaches like image splicing, retouching, copy-move, fusion-based, and deep learning-based is offered under one roof.
- A complete and consolidated list of information resources, journals, publicly available datasets, and performance parameters is provided.
- This study signifies concluding remarks for further research through a careful examination of the limitations and challenges of present studies.

1.4 Paper Organization

This paper is organized as follows: Sect. 2 addresses IFDT and their categorization using an appropriate hierarchy. Section 3 provides an outline of IFDT. Section 4 discusses the survey's review procedure, which includes research queries, sources of information, and publicly available datasets. A comprehensive review and analysis of IFDT are given in Sect. 5. Section 6 outlines the performance evaluation criteria. The challenges and future directions are offered in Sect. 7 and Sect. 8 includes the conclusion.

2 Taxonomy of Image Forgery Detection Techniques

This part demonstrates the taxonomy of IFDT based on a study conducted through numerous research papers. The taxonomy is created based on the approaches used for identifying picture forgery. Passive and active approaches are two basic kinds of IFDT. Active methodologies are subdivided into digital watermarking and digital signature procedures. The passive methodologies are further subdivided into geometry-based, physics-based, format-based, pixel-based, and camera-based approaches [1, 2, 21, 23, 24]. The taxonomy of IFDT is given in Fig. 3.

2.1 Active Techniques

The pre-processing procedures such as creating digital signatures or inserting watermarks into images are done during the image creation process in active IFDT. The elusive message concealed within a counterfeit picture is unretrievable. So, the validity of pictures may be confirmed in this manner [1, 21, 24].

2.1.1 Digital signature

This procedure extracts the image's characteristics while capturing the image. When the image is authenticated, the signature is produced again using a similar procedure. The uniqueness of picture can then be confirmed by comparison. To find the essential feature components, adaptive Harris corner and wavelet transform were used. The forensic signature was created using arithmetic feature modules based on neighborhood data. Fisher norm and Forensic signature were then utilized to determine the image's validity.

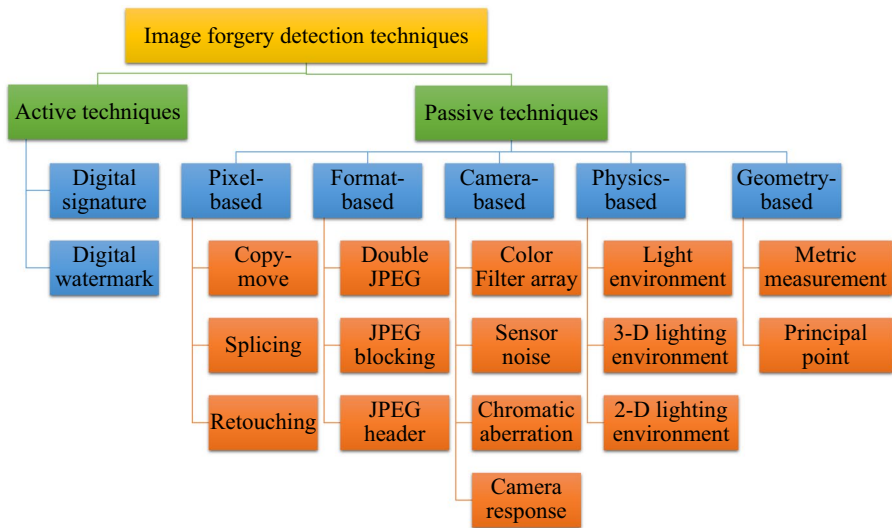


Fig. 3 Taxonomy of image forgery detection approaches

2.1.2 Digital watermark

A watermarked picture is produced when the source generates a watermark picture, which is then embedded into the picture itself. The watermarked picture is further used to recognize the watermark contained to confirm the image's validity. Development and testing of a computer-generated holographic (CGH) coding scheme for watermark inserting is described. The hologram was embedded using a blind additive embedding process.

2.2 Passive Techniques

The validity of the image is checked from the image itself in a passive IFDT. This technique, unlike active approaches, doesn't need any extra info, like signature or watermark [1, 2, 21, 23, 24]. The approaches for detecting passive image forgery are roughly grouped as follows: geometry-based, physics-based, format-based, pixel-based, and camera-based approaches. Figure 4 depicts the trends of publications indexed by the semantic scholar for passive approaches from 2006 to 2021. It is worth noting that pixel-based methods are still in their early phases of development in the last few years, which has grabbed the curiosity of many researchers. Consequently, this paper focuses on pixel-based approaches.

2.2.1 Pixel-based techniques

These methods identify counterfeit pictures by identifying statistical anomalies at a pixel level. Copy-move, image retouching, and image splicing are the three types of pixel-based methods. Figure 5 displays a graph from 2006 to 2021, indexed by semantic scholar, showing several articles versus three types of IFDT (copy-move, splicing, and retouching). The following are a few findings from this graph: In the previous few years, there has been a

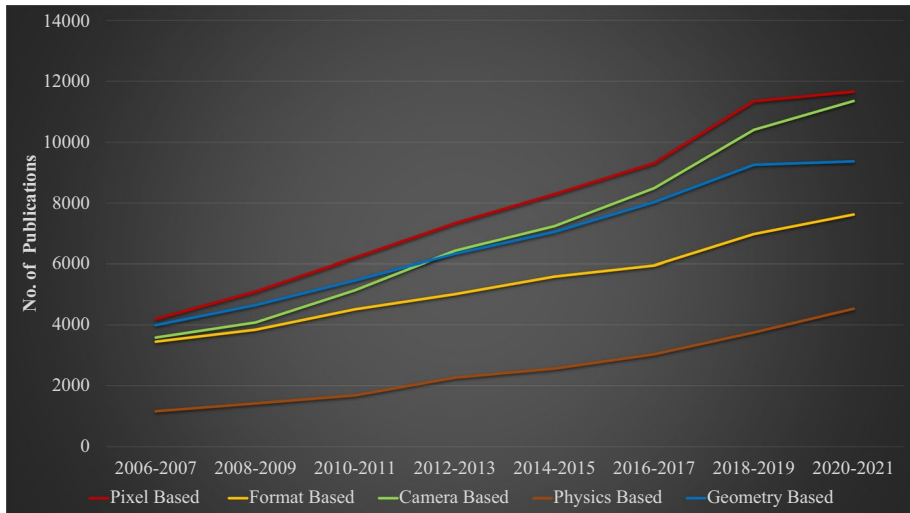


Fig. 4 Trends of publications related to passive IFDT

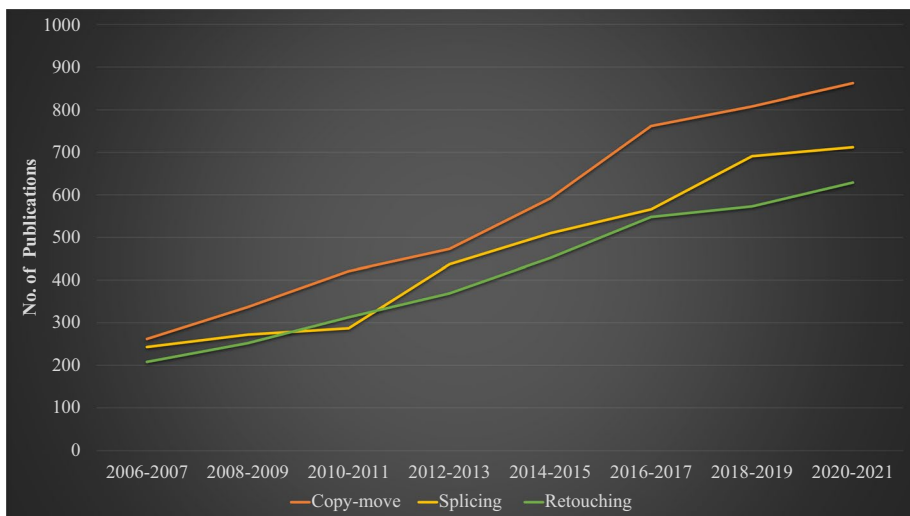


Fig. 5 Analysis of publications related to pixel-based methodologies

substantial increase in the detection of CMF, as well as a significant emphasis on image splicing detection in the recent decade. However, retouching detection receives less attention; one factor for this might be that it is the least harmful kind of forgery since retouched pictures are rarely misused for unlawful reasons.

Copy-move forgery It is a form of forgery. Assured areas of a picture are copied, moved, and inserted on a different section in a similar image using this approach. This type of forgery is used to obscure a specific section or replicate a section in a picture. The plain

CMF defines the act of duplicating a portion of a picture and placing it into another portion of the same picture without modifying it. In a multiple CMF, multiple areas or objects are duplicated and placed into different parts of a picture. In some cases, an item is duplicated from an image and placed in two separate locations. In a different scenario, several items are duplicated and inserted into various positions inside the same picture [21, 24, 28].

Image splicing It is a process for integrating multiple pictures. To some extent, splicing is comparable to CMF. The region copied in copy-move comes from the same picture, however, the area to be inserted in the spliced picture comes from a different picture. The splicing method involves replicating a specific section from one picture and placing it onto a different picture to create a fake picture. Image splicing involves compressing, resampling, and blurring areas to create a forged image. Consequently, spliced pictures are being exploited for nefarious reasons, because image splicing [21, 24].

Image retouching The technique of eliminating image defects, skin blemishes, scratches, and other imperfections is known as retouching. It is used to remove distracting aspects from photos by enhancing them according to the user's needs. Digital image processing provides a variety of retouching techniques and is considered suitable for modeling, beauty, and photo shoots. Illegal retouching may be used to cover the signs of forging operations that have already been performed on the image [21, 24]. Figure 6 shows examples of several types of pixel-based IFDT. The first and second column depicts original images and third and fourth column signifies forged images.

2.2.2 Format-based techniques

The statistical link generated by the lossy compression procedure is employed in format-based methods.

2.2.3 Camera-based techniques

Various image forensic techniques successfully exploit the abnormalities produced by the camera's sensor or lens in camera-based procedures. Based on various modeling ideas, these approaches are used to estimate the camera artifacts.

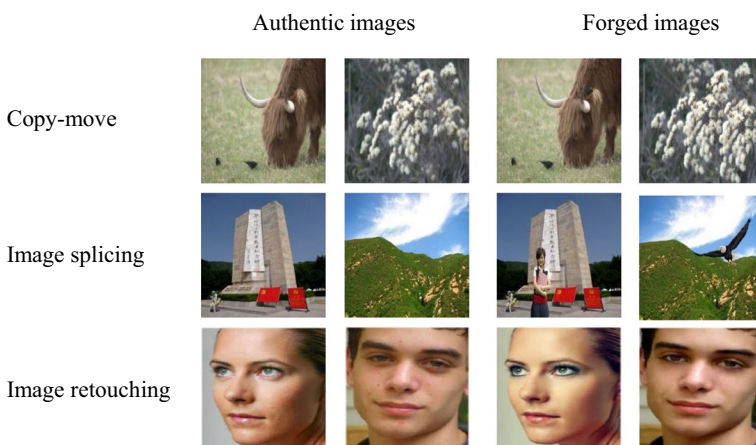


Fig. 6 Examples of different types of pixel-based IFDT

2.2.4 Physics-based techniques

Irregularities in the three-dimensional interface between the camera, light, and physical objects are clearly identified and modelled by these methods.

2.2.5 Geometry-based techniques

The variation in image's primary point works as proof to assess the image's validity in the geometry-based approach.

3 The Framework of Image Forgery Detection

The structure of IFDT is exemplified in Fig. 7.

- **Choosing an image dataset:** In digital forensics, choosing a dataset is crucial. A comprehensive dataset must include all sorts of forged images/videos that the classifier may face in reality. Many datasets are open to the public. Primarily, numerous studies were carried out on their datasets, nevertheless, maximum researchers eventually switched to using public datasets or synthetic versions. This has made comparing the performance of various approaches at ease.
- **Pre-processing:** To increase classification performance, a pre-processing step is performed. Many pre-processing approaches may be used before each image/video is subjected to feature extraction procedures to enhance the classification accuracy rate. Color conversion, equalization of JPEG quality factor, image blurring, noise filtering, and other methods are often employed.
- **Feature selection and extraction:** It is the act of converting a huge quantity of data into useful info. The classifier uses these characteristics to distinguish between original and tampered image classes. When choosing characteristics from an input image, factors such as the goal and approach of feature selection, feature dimensionality, and the computational cost of retrieving features are taken into account.
- **Classifier selection:** A classifier is a model-training method that takes a feature vector as input. A classifier creates a model after learning from training dataset. Support Vector Machine (SVM), Deep Learning, Decision Tree, Random Forests, Artificial Neural Network (ANN), and Naive Bayes are examples of classifiers. Accuracy, effectiveness, resilience, simplicity, and training time are all essential variables to consider when selecting a classifier.
- **Performance evaluation:** The performance of classifier is assessed using variety of parameters, including accuracy, training duration, sensitivity, specificity, precision, and so on. To examine the final classification tool, all of these characteristics are combined [1, 21, 23].



Fig. 7 Framework of image forgery detection

Figure 8 presents an overview of various features employed in IFDT. Here, two primary categories of these features are introduced i.e. methods based on spatial and transforms [29].

- **Spatial based:** Pixel location provides a direct representation of an image's content in a spatial feature space, characterized by an even distribution of energy and strong associations between neighboring pixels. The matching approach is consequently extraordinarily computationally challenging. Forgery in spatial feature spaces may be determined by intensities, moments, critical points, or textures [30]. Moment-based techniques like Hue, Zernike, and blur invariance moments may be utilized to evaluate IFDT. Images were partitioned into discrete sub-blocks to calculate their energies utilizing a range of intensity-based IFDTs. Key point-based methods concentrate the entire image on regions with high entropy. Various techniques have employed spatial-based approaches [31–37].
- **Transform based:** The correlations between transformed coefficients diminish, and a limited number of coefficients possess the majority of the energy. Due to this, the features utilized exclusively for overlapping blocks are these coefficients. There are three distinct types of transformations: frequencies, reduced dimensions, and texturing. A number of research articles have employed these techniques [29, 30, 38–40].

4 Review Procedure

The steps of this review method involve the development of a review framework, implementation of the survey, analysis of review outcomes, the recording of review results, and the exploration of research issues. The review approach employed in this

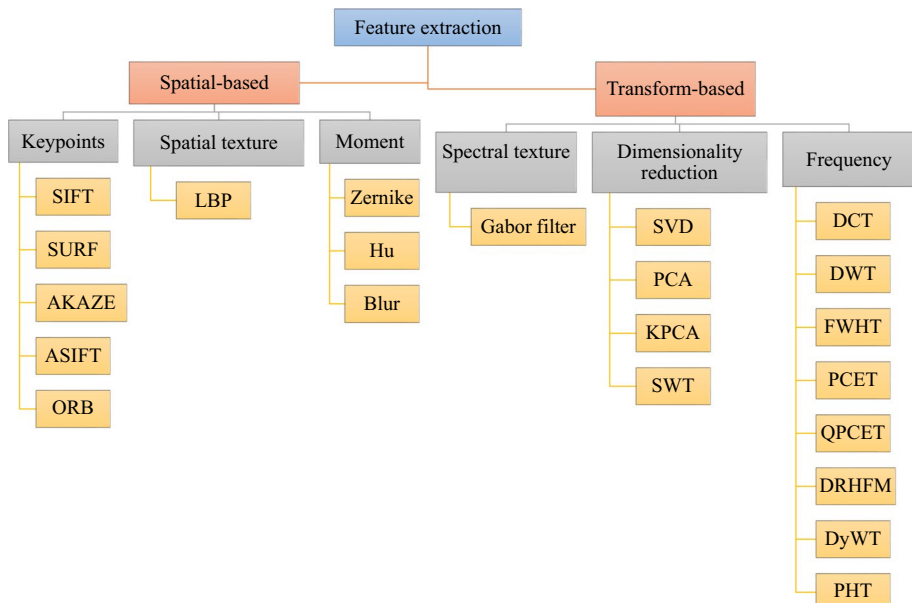


Fig. 8 Classification of feature extraction methods used in IFDT

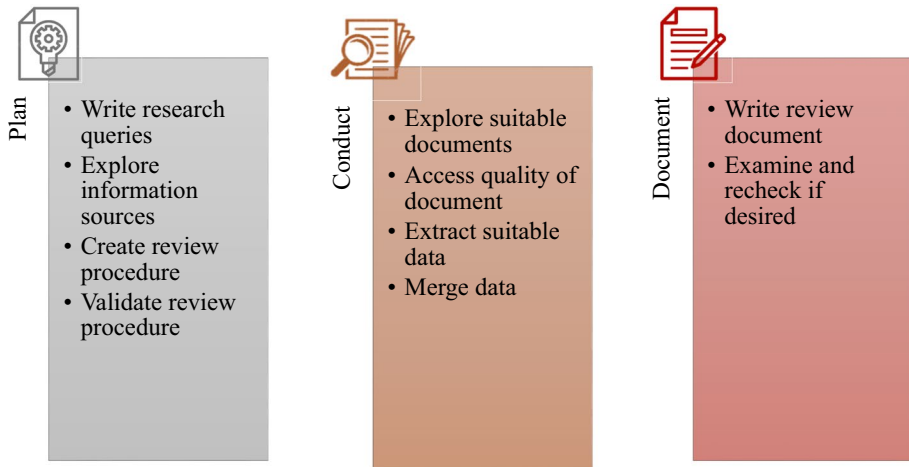


Fig. 9 Sections comprising the review process

Table 1 Research queries

Sr. no.	Research queries
RQ1	What are various categories of image forgery detection procedures?
RQ2	Describe the forensics taxonomy.
RQ3	What is the framework of image forgery detection procedures?
RQ4	What are the various methodologies used in the literature to support forgery detection techniques?
RQ5	What are the pros and cons of the existing methods?
RQ6	Describe various datasets used for image forgery detection.
RQ7	What is the trend of publications for passive image forgery detection techniques?
RQ8	Which journals are appropriate for publishing research articles on IFDT?
RQ9	What are the most well-known image forensics conferences/workshops/symposiums?
RQ10	How is the performance of image forgery detection procedures assessed?
RQ11	List a few sources of information and keywords used while searching for articles on image forgery.

systematic review is depicted in Fig. 9. Because the search policy is such a substantial part of the review procedure, the semantic scholar, google scholar, web of sciences has been enlisted to compile the necessary literature.

4.1 Research Queries

The focus is to conduct a thorough examination of the literature on IFDT. To accomplish this goal, numerous research queries are developed and responded to by analyzing the information extracted from the list of suitable research articles. Table 1 shows the research issues addressed in this study.

4.2 Information Sources

Public websites provide a broad source of information for academics to follow and distribute research articles. There is a various number of excellent research sites available, however, some of the most widely utilized sites for IFDT are listed in Fig. 10.

Figure 11 illustrates the keywords that are searched in reputable publications before finalizing the study’s content. Further, enormously regarded journals in field of image forensics are selected and shown in Fig. 12. These journals are the most valuable assets for scholars, which publish both reviews and research articles in area of image forensics.

In addition to journals, there are a few worldwide conferences in field of image forensics. Table 2 lists several Image Forensics Conferences/Workshops/Symposiums.

4.3 Description of Datasets

Because the area of image forensic research is continually evolving, there are limited openly accessible benchmarking datasets for study. These datasets must preferably include a mix of authentic and forged images. This offers a standardized platform for comparing various approaches. Table 3 describes the available datasets for forgery identification. The various datasets have been utilized based on the forgery type to be identified. Also, Fig. 13 illustrates the use of datasets indexed by Google Scholar during the previous years (2016–2024). According to this analysis, numerous datasets are used by different researchers, however, the Image manipulation dataset is used the most frequently.

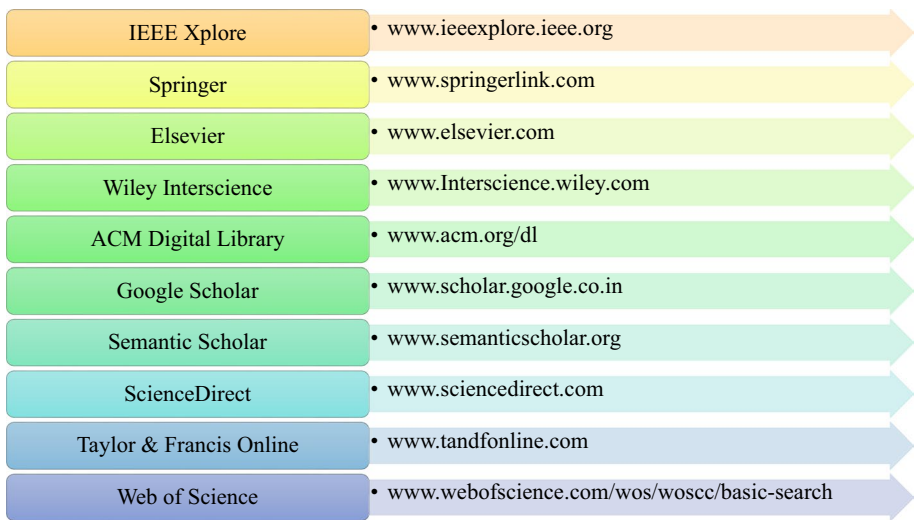


Fig. 10 Public websites for the source of information

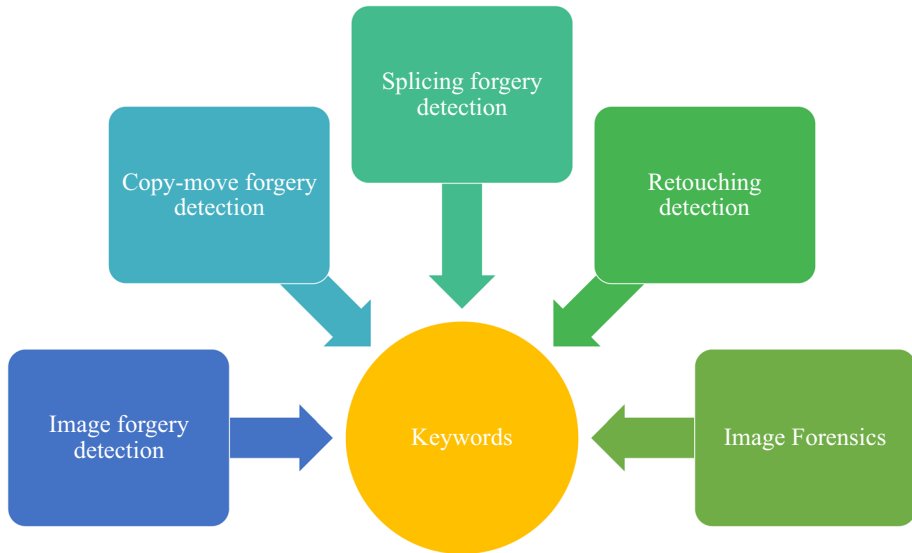


Fig. 11 Relevant keywords used in the search

Table 2 List of conferences/workshops/symposiums for image forensics

Sr. no.	Name of conferences/workshops/symposiums
1.	International Conference on Digital Forensics and Investigation (ICDFI)
2.	Conference on Computer Vision and Pattern Recognition (CVPR)
3.	International Workshop on Digital-forensics and Watermarking (IWDW)
4.	International Symposium on Digital Forensics and Security (ISDFS)
5.	International Conference on Electronic Security and Digital Forensics (ICESDF)
6.	International Workshop on Digital Watermarking
7.	International Symposium on Digital Forensic and Security (ISDFS)
8.	International Conference on Digital Forensics and Security of Cloud Computing (ICDFSCC)
9.	International Conference on Pattern Recognition (ICPR)
10.	IEEE International Symposium on Multimedia (ISM)

5 Review and Analysis of Passive Image Forgery Detection Techniques

Many researchers have been focusing their attention on the field of image forensics since the introduction of synthetic images, to identify different methods of image forgeries. Active and passive techniques are two types of IFDT. Active techniques examine the veracity of data to decide if a picture is manipulated with, such as digital signatures and watermarking. However, this sort of technology necessitates the use of specialized hardware or software to inject or retrieve genuine data from pictures before they are transmitted. The passive technique examines the image's contents and structure to determine its validity. It's noteworthy that among passive IFDT, pixel-based techniques are still in their early stages of development, which has grabbed the interest of many academics in previous years. As



Fig. 12 Useful journal list for detecting image forgery

a result, the focus of this study is on pixel-based passive IFDT such as image retouching, CMF, and image splicing forgery (ISF).

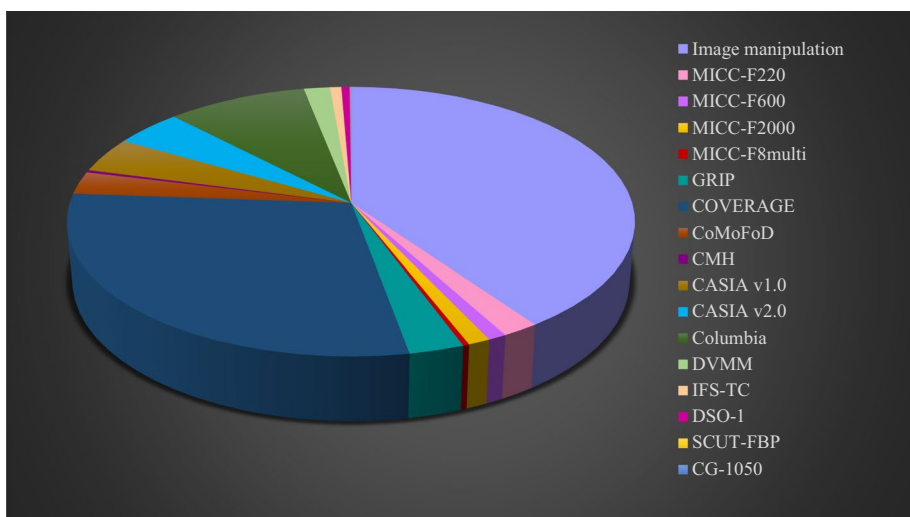
5.1 Copy-Move Forgery Detection Methodologies

The most common and well-known form of IFDT is CMF. To improve/eliminate characteristics, a specific portion of the picture is used. This system copies a segment of a photograph and places it into another domain within the same photograph to obscure particular characteristics in original picture or replicate data that is absent in original. Since CMF is performed in the same picture, transformed region's features will be identical to those of other portions. The CMFD methodologies are distributed into three methodologies as shown in Fig. 14. Block-based methodologies partition photographs into circular or rectangular blocks that are superimposed on top of one another and differ between methodologies. In contrast, keypoint-based methodologies extract characteristics from the entire

Table 3 List of available datasets relevant to forgery detection

Datasets	Images	File format	Size	Forgery type
Image manipulation * [41]	96	PNG	3000 × 2300	CMF
MICC-F220 * [42]	220	JPEG	722 × 480 to 800 × 600	CMF
MICC-F600 * [42]	600	JPEG, PNG	800 × 533 to 3888 × 2592	CMF
MICC-F8multi * [42]	8	JPEG	800 × 532 to 2048 × 1536	CMF
MICC-F2000 * [39]	2000	JPEG	2048 × 1536	CMF
GRIP * [43]	160	PNG	768 × 1024	CMF
COVERAGE* [44]	200	TIF	400 × 486	CMF
CoMoFoD * [45]	1040	JPEG, PNG	512 × 512	CMF
CMH * [46]	216	PNG	845 × 634 to 1296 × 972	CMF
CASIA v1.0 * [47]	1721	JPEG	384 × 256	CMF and ISF
CASIA v2.0 * [47]	12,614	JPEG, TIFF, BMP	240 × 160 to 900 × 600	CMF and ISF
Columbia * [48]	363	TIFF, BMP	757 × 568 to 1152 × 768	ISF
DVMM* [49]	1845	BMP	128 × 128	ISF
IFS-TC * [50]	2200	PNG	1024 × 768 to 2848 × 2144	ISF
DSO-1 * [51]	200	PNG	2048 × 1536	ISF
SCUT-FBP * [52]	500	JPEG	384 × 512	Retouching
CG-1050 * [53]	1150	PNG	3456 × 4608/4608 × 3456	CMF, ISF and retouching

* indicates link of datasets

**Fig. 13** Usage of datasets in last few years (2016–2024)

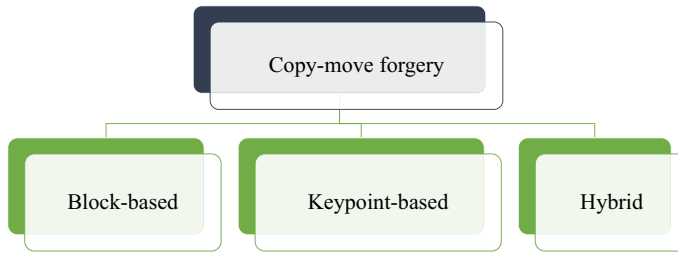


Fig. 14 Classification of CMFD methodologies

picture [1, 2, 22, 41, 42, 54]. Hybrid procedures are the fusion of both keypoint-based, and block-based procedures. Figure 15 shows an illustration of CMF.

5.1.1 Block-based techniques

These procedures extract attributes from a single block and then compare them to attributes from various blocks. If the attributes of two blocks are same, block is inspected to confirm CMF. Many existing block-based methodologies include Zernike moments, Fractional Quaternion Zernike Moments (FQZM), discrete cosine transform (DCT), Cellular Automata (CA), Discrete Radial Harmonic Fourier Moments (DRHFM), discrete wavelet transform (DWT), and many more. Table 4 shows several block-based CMFD methods.

The DCT-based approach for CMFD was initially proposed by Fridrich et al. [55]. At bitmap graphics, the picture is split in fixed-size overlaying picture sections, and DCT is employed for every section. The quantization feature vector is produced by examining the quantized DCT coefficient matrix in a zigzag pattern. The feature matrix is lexicographically structured, and match is determined via Euclidean distance. This procedure, however, has a large computational cost.

Popescu et al. [38] reduced the dimensionality of block characteristics by principal component analysis (PCA) and presented methodology for detecting counterfeit domains in forged images. This format is unaffected by slight image variations caused by additive noise or lossy compression. However, it fails to execute image rotation, resizing, and

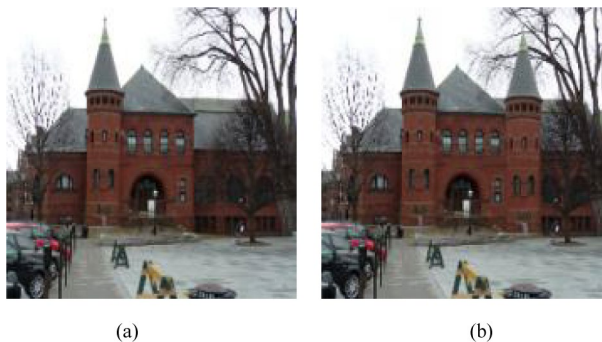


Fig. 15 An example of CMF (**a** genuine picture with single pillar; **b** counterfeit picture where a pillar on left side of a building is copied and pasted on right side)

Table 4 Block-based CMFD techniques comparison

Reference	Dataset used	Pre-processing	Feature selection and extraction	Merits	Demerits
Cao [65]	DVMM, Kodak pictures, internet pictures	RGB into grayscale, circular block division	DCT	Robust against multiple CMF, noise addition, Gaussian blurring	Ineffective against rotation, scaling, JPEG compression
Zhong [66]	MICC-F220, MICC-F600, MICC-F2000, MICC-F8multi	RGB into grayscale, Gaussian filter, Circular block	DRHFM	Perform better under rotation, Gaussian noise, and JPEG compression	Ineffective against multiple CMF
Wang [67]	MICC-F220, MICC-F2000	RGB into grayscale,	DCT, Perceptual Hashing	Robust for attacks like contrast ratio, luminance, hue, Gaussian noise, and blurring	Unsatisfactory time complexity, non-resistant against rotation and scaling
Chen [68]	IMD, GRIP	Circular block division	FQZM	Perform well for scaling, and rotation operation	Not robust to other additional operations
Hosny [69]	Internet pictures	RGB to HSV, Median filter	PCET	Robust against JPEG compression, rotation, Gaussian noise, and scaling	Unable to detect CMF in color images
Meena [70]	CoMoFoD, GRIP, CMF	RGB into grayscale, overlapping block division	Gaussian-Hermite moments	Able to detect small portion, robust against blurring, rotation, scaling, JPEG compression	Unable to perform deep learning
Nirmal Jothi [71]	MICC-F220, MICC-F2000, UCID, CoMoFoD, CASIA	RGB into grayscale, Haar wavelet transform	Local Binary Pattern Variance, HOG descriptor	Robust against rotation, blurring, flipping, etc., better time complexity	Not able to perform on video and deep learning
Gani [72]	CoMoFoD	RGB into grayscale, block division	DCT, CA	Robust against noise and compression	Large time complexity

flipping operations. To identify unique artifact, Li et al. [56] developed blind forensics technique based on DWT and SVD. To enhance localization of the copied area, Bravo-Solorio et al. [57] generated a 1D descriptor that is uniform to reflection and rotation by aggregating a log-map across the log radii axis. Yap et al. [40] developed a technique against rotation based on polar harmonic transforms (PHTs). Bravo-Solorio et al. [58] detected counterfeit regions that have been affected by scaling, reflection, and rotation. However, it has a higher computational complexity.

Wu et al. [59] utilized log-polar fast Fourier transform (LPFFT) to identify forgeries even when the counterfeit region is rotated and/or scaled. It is capable of producing acceptable accuracy while using significantly lesser computation complexity. Muhammad et al. [60] propose a blind CMFD procedure depending on undecimated dyadic wavelet transform (DyWT). This technique is tested in three scenarios: constant size pictures and counterfeit without rotation, varied size pictures and counterfeit with or without rotation, and varying quality factors of JPEG images. Zhao et al. [8] offered robust procedure based on SVD and DCT to identify CMF. This methodology can identify multiple CMFs and correctly spot duplicated portions even when a picture has been affected by Gaussian blurring, AWGN, JPEG compression, or their combined processes. However, it fails against rotation and scaling attacks. Further, PatchMatch, a fast approximate nearest-neighbor search methodology designed specifically for computing of dense areas over images, is used by Cozzolino et al. [43] and Zhong et al. [61] introduced a unique approach for supporting CMFD using discrete analytical Fourier- Mellin transform (DAFMT).

Mahmood et al. [62] use both the stationary wavelet transform (SWT) and DCT to expose CMF in digital images. The suggested approach initially employs SWT on the picture to obtain approximation subband. Following that, approximation subband is split into fixed-length overlying blocks. The experimental findings show that it is resistant to different geometrical assaults. Ouyang et al. [34] identify CMF, although it takes longer to run. Meena et al. [63] discovered tampered regions employing the tetrolet transform in case of numerous post-processing processes, however, performance factors still need to be improved. Using DCT and SVD, Priyanka et al. [64] identified and located CMF. This technique yields high recall results; though, it offers a lower accuracy value.

Al-Qershi et al. [73] suggested a better matching methodology based on K-means clustering technique. Furthermore, the matching approach is used in this algorithm combined locality-sensitive hashing (LSH) with Zernike moments. Although the suggested technique improves detection accuracy and decreases processing time, it has a lower precision and recall value. Kasban et al. [74] extracted Hilbert-Huang Transform (HHT) characteristics from Cr component after first converting the RGB image to YCbCr space. To determine counterfeit detection accuracy, the conclusions are validated using Structural-Similarity (SSIM). In addition, the suggested method has been evaluated against a variety of post-processing threats. A novel CMFD approach based on Kolmogorov Smirnov (KS) test and SVD was developed by Ahmed et al. [75]. Retrieved characteristics are lexicographically sorted and the KS test is used to match them. It is resistant to color alteration, brightness change, picture blurring, and contrast modification.

Babu et al. [76] offer a two-step method for detecting forgeries. Optimized SVM is utilized to classify the images. If the suspected picture is classified as fake, then it is transformed into overlying blocks. The suggested method needs to be modified in the future to evaluate if an image is legitimate or produced by a machine. Gani et al. [77] present a block-based CMFD method that performs well with post-processing modifications. In addition, a modest and quick CA-based technique is provided to reliably recover the cloned areas matching the matched characteristics. However, it has a long feature extraction time.

The main disadvantage of block-based approaches has been identified as their computational cost, and more powerful feature extraction algorithms are required.

5.1.2 Keypoint-based techniques

This technique is depend on Polar Complex Exponential Transform (PCET), Accelerated KAZE (AKAZE), Speeded-up Robust Features (SURF), Scale-invariant Feature Transform (SIFT), Affine Scale-invariant Feature Transform (ASIFT), Oriented FAST and Rotated BRIEF (ORB), etc. Table 5 shows a comparison of several keypoint-based methods.

Pan et al. [78] pioneered the use of key point matching for CMFD. Although this technique performs well against geometric alterations when the parameters are calculated by Random Sample Consensus (RANSAC) procedure, a more vigorous procedure is necessary. Based on SIFT characteristics, a novel methodology for image forensics analysis has been presented by Amerini et al. [42]. The described approach is successful in a variety of operating circumstances, including composite processing and multiple duplications. Zhao et al. [79] propose a passive forensic approach for identifying region duplicate picture counterfeiting depending on local binary patterns (LBP) and Harris feature points. Experimental results indicate that the procedure can identify CMF even under geometrical assaults and that it is extremely resistant to forging with the smooth area of small structures. Despite its promising detection performance, the suggested approach fails to identify CMF with scaling since LBP and Harris corners are susceptible to picture scaling, and thus, they are not being calculated on multiscale images.

For identifying identical areas in an image, Jaberri et al. [80] used mirror reflection invariant feature transform (MIIFT). The outcomes show that the technique can identify CMF with greater accuracy, specifically when the size of the identical portion is small. Yu et al. [81] present a feature-based system to improve feature coverage and matching efficiency. This approach can withstand modest degrees of scaling, additive noise, and their combination, nevertheless, performance suffers fast when these assaults are significant, owing to Harris Corner Detector's instability in such situations.

Yang et al. [82] suggested CMFD system based on hybrid characteristics. To retrieve additional feature points, KAZE, a strong interest point detector, is employed and coupled with SIFT. Experiments demonstrate that this approach can efficiently detect duplications, even when the duplicated parts have experienced multiple geometrical modifications. An improved keypoint-based CMFD for tiny smooth areas is introduced by Wang et al. [83]. The suggested CMFD technique has a greater computational cost, which means it cannot be employed successfully in real-time applications.

Alberry et al. [84] offered a unique strategy to utilize Fuzzy C-Means (FCM) clustering throughout the matching procedure. Before actual matching phase, the method employs FCM to gather the observed SIFT key points. Consequently, computational cost is reduced, resulting in a faster matching process. Wang et al. [85] discovered and localized CMFs depending on color resemblance framework. Further, pairs of linked Delaunay triangles are constructed, and appropriate local graphic characteristics, QPCET coefficients are calculated.

Dhivya et al. [35] employ SURF and SVM. The SURF algorithm picks most prominent spots in the input images. The SVM classifies the incoming images using object recognition to retrieve the matching feature points. By reducing the contrast threshold and scaling picture, Li et al. [36] presented an acceptable number of key points in smooth domains, but it has a long computing time. Yang et al. [37] offered a CMFD approach depending

Table 5 Keypoint-based CMFD techniques comparison

Reference	Dataset used	Pre-processing	Feature selection and extraction	Merits	Demerits
Lyu [91]	IMD, MICC-F600	Not applicable	LJOP keypoints extraction from Delaunay triangles	High recall and acceptable precision	Not robust against some geometrical attacks
Chen [92]	IMD, GRIP, Dataset	Not applicable	SIFT	Precisely detect tampered locations for simple, geometric transformation, and small-scale forgery	Shows poor effect in large-scale forgery
Bilal [93]	MICC-F220, CoMoFoD, MICC-F2000	RGB into grayscale, dynamic histogram equalization	SURF	Attain better value of performance parameters	More difficult in compression attack
Wang [94]	GRIP, IMD, SBU-CM16	Minimum barrier superpixel segmentation	SURF, PCET	Can endure attacks like scaling, blurring, noise addition, etc	Difficult to detect large-scale and small-scale forged areas in images
Prakash [95]	IMD	Image resizing, Convert image into HSV color space	AKAZE, SIFT	Robust under rotation, noise addition, plain and multiple CMF	The performance parameters need to be enhanced
Wang [96]	IMD, GRIP	Not applicable	A KAZE, SURF	Able to detect CMF under image modifications	It is a time-consuming approach
Ardizzone [97]	CVIP	Not applicable	Harris Corner Points, SURF, SIFT	High precision value and is resistant to geometric changes	Missing patches cannot be retrieved and need to improve recall value

on adaptive keypoint removal and matching. Initially, adaptable even distribution criteria are used to extract image key points. Next, to express the local picture characteristic of picture keypoints, the BRISK descriptor is designed. Then, using the embedded random ferns method, local BRISK attributes are used to correlate picture key points. Lastly, the RANSAC is used to remove erroneously matched keypoint pairs, and tampering regions are localized.

Liu et al. [86] integrated Local Intensity Order Pattern (LIOP) and SIFT keypoints to exploit supremacy of various key points. Wang et al. [87] suggested an efficient and high image CMFD framework that incorporates obtaining adaptive image keypoints, presenting a fast robust invariant feature, and sorting out incorrect pairs. The smoothed keypoints are recovered dynamically from the counterfeit image using the rapid approximated Laplacian of Gaussian (LoG) filter.

The CMFD technique is suggested by Uma et al. [88] and only considers a small subset of the strongest key points (KPs). Niu et al. [89] presented a quick and efficient CMFD. Primarily, a BFM-based descriptor is intended to provide a quick, robust, and exclusionary presentation of dense key points. The total matching complexity is then lowered by constructing a hierarchical matching method based on BFM invariant magnitude and phase data. Yang et al. [90] present Two-Stage Filtering (TSF), an effective and economical CMFD method. This technique is used to filter out key point pairs that do not match. Although it is superior at identifying forgeries when compared to post-processing and large-scale scaling procedures, it has a long average detection time.

The primary benefits of keypoint-based CMFD techniques are their remarkable performance in terms of computing cost, memory requirements, and uniform scaling and rotation procedures, although they indeed struggle to decrease false matches in plane areas. Another issue with key-point-based techniques is that in smooth/flat and homogenous regions, fewer key points are extracted.

5.1.3 Hybrid techniques

Each keypoint-based technique has its own set of advantages and disadvantages. Duplication detection can sometimes be improved by using a mix of various keypoint-based techniques. Some researchers have also suggested combining keypoint-based approaches with block-based methodologies to address the deficiencies of keypoint-based methodologies in flat regions. As a result, several features of block-based methods are merged with classic keypoint-based approaches to generate hybrid techniques.

In their research on IMD, Pun et al. [98] employed feature point matching and adaptive over-segmentation. For successful CMFD, Zernike moments and SIFT are implemented in these two types of areas, smooth and non-smooth by Zheng et al. [99]. Sun et al. [100] propose a CMFD approach capable of addressing the problem of computation cost. Using SIFT and Zernike moments, features are retrieved and matched in the textural and smooth regions, respectively. Since it is resistant to various attacks, however, the impact of the process is not very excellent while dealing with highly smooth blocks, like walls and sky. So, there is a need to enhance location accuracy and detecting speed.

By hybridizing DCT and SURF, Ojeniyi et al. [101] discovered the CMF. This technique is capable of addressing some of the challenges concerned with CMF; however, there is still a need for enhancement in detecting other types of image formats in jpeg format. Depending on SLIC segmentation and Helmert transformation, Huang et al. [102] employed Helmert transformation to compute geometric features among matched pairs and

to deal with merging clusters. A SLIC algorithm, on the other hand, is utilized to more accurately locate the tampering locations.

Over-segmentation is employed as a preprocessing step by Elhaminia et al. [103] to generate superpixels, which are subsequently treated as neurons in the Markov network. The SLIC technique is used to split pictures into multiple superpixels, and then PCT and SURF characteristics are utilized for matching features and identifying comparable portions of picture. Further, SLIC superpixel segmentation and K-means clustering are used by Liu et al. [104] to split a picture into complicated areas. To identify tampering in complicated regions, SIFT characteristics are utilized. To identify alterations in even zones, segment mask characteristics and RGB color characteristics are offered. Nonetheless, a more precise approach is required.

Niyishaka et al. [105] offer an efficient approach for detecting CMF that makes use of picture blobs and BRISK attribute. This approach has a large computational cost. Meena et al. [106] recommended a hybrid methodology that combined Fourier-Mellin Transform (FMT) with SIFT. The key points are retrieved from texture component of the picture with SIFT descriptor, and FMT is employed to the image's smooth portion. The retrieved characteristics are compared to find the image's replicated areas. The simulation findings show that this approach outperforms in terms of geometric alterations and post-processing procedures in an acceptable amount of time.

Agarwal et al. [107] offer a robust CMFD system employing hybrid CMFD approaches. Using an enhanced FCM clustering system relying on superpixels, a photo is separated into non-overlapping blocks. Although it achieves outstanding forgery detection results even in the most challenging conditions, still there is a need for an enhanced method using deep learning.

Tinnathi et al. [108] developed a unique CMFD procedure that makes use of adaptive segmentation and a hybrid feature extraction system. The tampered picture is adaptively segmented into non-overlapping parts, which improves efficiency of the CMFD method and reduces computing complexity. When the image suffered certain geometrical changes, the use of Hybrid Wavelet Hadamard Transform (HWHT) to retrieve the segment characteristics, enhanced the efficacy and gave a robust result. Tahaoglu et al. [109] used contrast-limited adaptive histogram equalization (CLAHE) on three independent channels from two color spaces (RGB and $L^*a^*b^*$) to extract SIFT keypoints.

5.2 Image Splicing Forgery Detection Techniques

Another common method for creating the forgery is image splicing. It comprises exchanging segments of one or more areas of the picture with segments of a different image. Splicing is a system of picture modification in which parts of the same or distinct pictures are combined to form a counterfeit image without any additional processing such as smoothing and blurring of borders and edges between fragments. ISF can cause differences in a variety of ways, such as an abnormal sharpness at the borders, which can be employed to detect manipulation. Figure 16 depicts the illustration of image splicing forgery (ISF), in which two original pictures are joined to make a counterfeit picture. When two separate pictures captured at different times and locations are combined, the original and counterfeit pictures become unbalanced; this discrepancy is used as a criterion for identifying ISF. Table 6 compares numerous ISF detection strategies.

Ng et al. [110] presented an ISF detection technique that makes use of bicoherence magnitude and phase characteristics. Shi et al. [111] regarded an image's adjacent



Fig. 16 An illustration of ISF **a** genuine picture of mountains and sky, **b** genuine picture of a bird in the sky, **c** spliced image with the combination of **(a)** and **(b)**

changes in BDCT parameters as 1-D signal. The relationships among nearby nodes in a specific orientation (vertical or horizontal) are represented by a Markov process and transition probability matrix (TPM) that is utilized as an exclusionary feature vector for SVM classifier. SVM classifier is applied by Li et al. [112] on HHT-based and moment features.

He et al. [113] combined the Markov characteristics in the DWT and DCT domains. Even though the article established the legality of the Markov feature, the overall accuracy still has to be improved. Rao et al. [114] used motion blur signals to reveal ISF. Carvalho et al. [51] utilized an SVM meta-fusion classifier that is based on color differences in image illumination.

LBP and steerable pyramid transform (SPT) are employed to identify image splicing by Muhammad et al. [115]. The colored picture is first converted to YCbCr, then SPT transform is applied to Cb and Cr channels. LBP histograms are then used to characterize the texture in each SPT subband and SVM classifies images as fake or original using the feature vector. Pun et al. [116] uses noise differences at different scales as indications for ISF detection. Each scale computes a noise level job that returns the relationship between the noise level and illumination of every fragment. Fragments that are not limited by noise-level jobs are considered mistrustful. After the appropriate morphological processing, pixels appearing in mistrustful areas of every scale are designated as a spliced area in the final phase.

To identify image splicing, a Markov model-based methodology is suggested by Zhang et al. [117]. Markov model is used in block DCT and Contourlet transform domain. Furthermore, in transform domain, new attributes are retrieved to define the dependence of locations among Contourlet subband parameters. Ultimately, for grayscale images, SVM is used to identify genuine and spliced pictures, whereas an ensemble classifier is used for the color pictures. Zhao et al. [118] propose a 2-D Markov scheme for ISF detection. The approach is complicated, but it is more resistant to JPEG compression and median filtering.

Chen et al. [119] consider both RGB and depth (RGB-D) information while improving quaternion representation (QR) for effectively expressing RGB-D images. Experiment findings on datasets illustrate that the proposed ISF detection system attains good accuracy with the proper feature dimensionality. El-Alfy et al. [120] provide a reliable technique for substantiation of grayscale and color pictures. The global and local objects caused by image modification are captured by integrating Markov attributes in DCT and LBP domains. The combination of dimension reduction utilizing PCA and an optimized SVM classifier has proven to be effective. To begin, color information is recovered by Li et al. [121] from blocked pictures to create a quaternion in its entirety, and QDCT coefficients of quaternion blocked pictures are acquired. Second, the extended Markov characteristics derived

Table 6 Comparison of ISFD methodologies

Reference	Dataset used	Pre-processing	Feature selection and extraction	Classifier	Merits	Demerits
Alahmadi [128]	CASIA v2.0, Columbia	RGB to YCbCr, Overlapping blocks	DCT, LBP	SVM	Provides good accuracy	Unable to localize spiced regions
Sheng [129]	CASIA v2.0,	RGB to Grayscale, Block division	Discrete Octonion Cosine Transform	LIBSVM	Robust against blurring and Gaussian noise	Large algorithm's complexity
Zhang [130]	DVMM, IFS-TC, NIKON, GrayNIKON	Not applicable	Markov attributes in block DWT domain	SVM	Suited for dealing with datasets generated by JPEG compression	Incapable to process large image data
Pham [131]	CASIA v2.0	Not applicable	Coefficient-wise and block-wise Markov features in DCT domain	SVM	Decrease total running time and increase detection accuracy	Optimization problem between accuracy and computational complexity
Jaiswal [132]	CASIA v2.0, Columbia, IFS-TC	RGB to Grayscale	DWT, Fourth-order central moment	NA	Can localize spiced regions	Do not provide a good value of performance metrics
Kanwal [133]	CASIA v1.0, Columbia,	RGB to YCbCr, Block division	Otsu Based Enhanced Local Ternary Pattern (OELTP)	SVM	Useful in detecting multi-image fraud even in formats such as jpeg	Spiced areas cannot be localized
Siddiqi [134]	CASIA v2.0, Columbia,	RGB to YCbCr, Block division	DWT, Dominant Rotated Local Binary Patterns (DRLBP)	SVM	Identify ISF even in the existence of post-processing operations	Unable to localize spiced regions

from transition possibility arrays in the QDCT domain preserve relationship between block QDCT coefficients. Subsequently, SVM is used to categorize the Markov feature vector.

An effective SVD-based feature extraction approach for ISD is presented by Moghadasi et al. [122]. The method employs several methodologies, including roughness measure procedure to proficiently estimate the set of acquired singular values; kernel PCA system employed as a feature reduction phase to enhance classification. Using DCT coefficient quantization, Han et al. [123] present an effective Markov feature extraction approach for ISF detection (ISFD). The quantization process improved ISFD's performance. Furthermore, two Markov techniques based on summary and maximum of directional and color characteristics are discussed.

Subramaniam et al. [124] produce DWT coefficients for enhancing ISF detection. The methodology alters the content of a manipulated picture and produces abnormalities in picture characteristics. To evade identification, the spliced region's borders are generally blurred. So, FMOs and CFMs are utilized to determine amount of hiding of altered area limits for ISF. Jalab et al. [125] identified ISF using approximated Machado fractional entropy (AMFE). Although it produces superior outcomes, it is affected by post-processing procedures alike JPEG compression. Kaur et al. [126] extracted and combined Markov attributes from both LBP and DWT domains to identify ISF. The suggested scheme's efficacy is evaluated using six benchmark datasets, and SVM classifier categorized the pictures as altered or genuine.

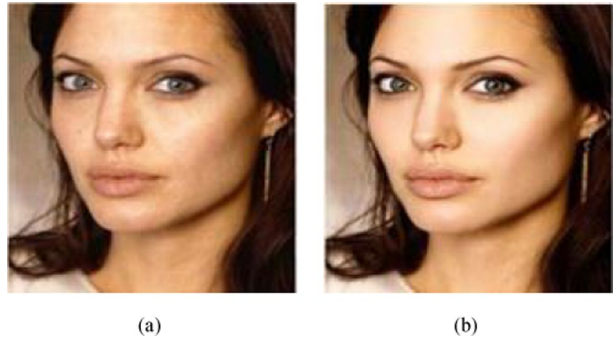
Niyishaka et al. [127] present a simple and highly efficient ISF detection method that takes into account a trade-off among requirements and user cost. The phases in the technique are as follows: Initially, chrominance and luminance are extracted from the source images; and lastly, LBP normalized histogram is calculated. Although the suggested approach is computationally cheap and reliable for ISF detection, it cannot locate spliced images. Itier et al. [32] suggest detecting and localizing ISF by evaluating the connection of image noise among R, G, and B channels that is an inherent part of digital cinematography implementation. To be more specific, the scheme detected the border between the host picture and spliced portion.

5.3 Image Retouching Forgery Detection Techniques

Image retouching is essentially a tool for forging images that may be used for both viable and creative purposes. Retouching is a technique for enhancing or reducing the importance of certain aspects of an image. Color correction, glamour correction, skin retouching, image reconstruction, picture cartooning, and so on are all examples of retouching. The image does not change dramatically in image retouching, but it improves or lowers specific characteristics of the image, which is extremely common in most photo editing software, as illustrated in Fig. 17. In most picture magazines, there is a need for image attractiveness, which leads to the enhancement of some features of an image, even though such a method is illegal.

Stamm et al. [135] investigated a strategy that identifies global changes and recommends histogram equalization techniques. Stamm et al. [136] suggested an iterative methodology to mutually measure any arbitrary contrast enhancement mapping. It is required to alter a picture and pixel value histogram of a picture before contrast enhancement. The probabilistic form of a histogram of the picture is used to conclude which entries of histogram mainly resemble contrast enhancement artifacts. Simulation results specify that the

Fig. 17 Illustration of image retouching forgery **a** original picture, **b** retouched picture



method is capable to give precise guesses even as nonstandard types of contrast enhancement are applied to the image.

Cao et al. [137] introduced a novel technique for identifying unsharp masking sharpening depending on feature overrun abnormalities at side-planar edges. As per the results of an experimental investigation, the authors' approach is effective in detecting sharpening on small pictures, though post-JPEG compression and noising assaults are used. Ding et al. [138] presented a scheme for detecting the unique texture alteration induced by USM sharpening. Cao et al. [31] introduced two algorithms to discover contrast enhancement engaged manipulations in digital imagery. At the outset, the identification of global contrast enhancement is the main spotlight that has been employed to formerly JPEG-compressed pictures. Furthermore, it has been introduced to discover the complex pictures produced by implementing contrast modification on both or any one of the source areas. In order of determining the image forgeries as well as placing the composition limits, the reliability among regional artifacts has been analyzed. The widespread research has documented the efficiency and usefulness of the planned methods.

Zhu et al. [139] detected picture sharpening operations using the overshoot artifact measure. The approach provided by Vazquez-Padin et al. [140] employed Random Matrix Theory (RMT). RMT simulates the action of eigenvectors and singular points in any matrix. The computation cost of RMT is quite low.

5.4 Fusion Based Techniques

In fusion-based techniques, various forensic techniques are integrated to increase the reliability of procedures. CMF and ISF are targeted by these approaches in the majority of recent research. Liu et al. [141] focused on discovering the two most common kinds of image alteration, copy-move, and splicing forgeries. A single authentication procedure is used to locate forged areas using an integrated methodology. JPEG block synthetic networks and local noise inconsistencies are utilized to produce attributes in this approach, that are then merged with the picture quality score as a factor. Prakash et al. [142] used a block DCT (BDCT) and an improved threshold technique to extract features. Jaiprakash et al. [143] presented a low-dimensional feature-based approach for detecting CMF and ISF. Image characteristics and pixel correlations are used to extract features from the DCT and DWT domains.

Dua et al. [144] offer a complete technique for evaluating JPEG compressed tampered pictures. With remarkable detection accuracy, forged pictures are detected from a standard

collection of pictures using a double stochastic model with quantized DCT coefficients. Pham et al. [145] present a robust technique for clustering relevant pictures. The suggested model finds a cluster center that is the sole legitimate picture in a cluster, from picture clusters by leveraging structural correlation among images. Kaur et al. [146] offer a combined model employing LBP and discrete fractional cosine transform (DFrCT) to identify CMF and ISF simultaneously. The DFrCT's fractional parameter is used to improve accuracy, while LBP is used to effectively identify tampering artifacts. The pictures are then classified using SVM.

5.5 Deep Learning-Based Techniques

Deep learning has grown ambitiously for image forensics with the advent of methods in artificial intelligence. Training samples are utilized to retrieve attributes in various deep learning configurations, including Convolutional Neural Network (CNN) and deep residual architecture. So, several procedures are proposed to address image forgeries in recent times. Table 7 compares several deep learning-based strategies.

AL_Azrak et al. [147] developed structure for identifying CMF. CNN is employed to detect forgeries. To differentiate between real and manipulated images, Goel et al. [33] employed a dual branch CNN. This approach is vulnerable to many assaults. A modified CNN was used by Rao et al. [148] as a patch descriptor to identify both CMF and ISF. Xiao et al. [149] introduced a technique for identifying ISF that consists of two parts: a coarse-to-fine CNN (C2RNet) and diluted adaptive grouping. The suggested C2RNet retrieves changes in picture characteristics among genuine and modified areas from picture patches of different sizes.

Jaiswal et al. [150] created a CNN model that uses multi-scale input. In the encoder block, retrieved feature maps from several phases of convolutional layers are merged and downsampled. The resulting feature map is utilized to categorize the images using a sigmoid activation function. Rao et al. [151] suggested a unique ISF detection and localization technique based on a deep CNN-learned local feature descriptor. A structure based on a custom design and transfer learning are the two deep learning strategies that Rodriguez-Ortega et al. [152] recommended. The metrics achieved by the VGG-16 architecture using transfer learning are 10% higher than those of the framework using bespoke architecture, however the inference time is almost twice as long.

6 Performance Evaluation Criterion

The performance evaluation metrics such as detection accuracy (DA), Mathews correlation coefficient (MCC), recall (R), area under curve (AUC), F_1 score (F_1), markedness, precision (P), receiver operating characteristics (ROC), F_2 score (F_2), informedness, and others are used to determine the effectiveness of the detection approaches based on the confusion matrix shown in Fig. 18. Positive and negative predictions for True Positive (T_{PO}), False Positive (F_{PO}), True Negative (T_{NE}), and False Negative (F_{NE}) are included in the confusion matrix. In general, T_{PO} instances are ones in which both the predicted and actual results are accurate. The F_{PO} instances are ones in which the predicted outcome is true but the actual outcome is false. The T_{NE} instances are those in which both the predicted and actual outcomes are incorrect. Predicted outcomes are false in F_{NE} instances, whereas actual results are real. In image forgery detection, T_{PO} is the total images impeccably recognized as counterfeit, F_{PO} exemplifies the

Table 7 Deep learning based techniques comparison

Reference	Dataset used	Forgery type	Feature extraction	Model	Merits	Demerits
Chen [153]	Collected from different cameras	Filtering, Cut and Paste	Median filter residuals	CNN	Good accuracy	Need to test on more images
Liu [154]	MFC NIST 2018, CASIA v2.0, Reditt, PS-Battles	Image splicing	Atrous convolution	Deep matching network based on atrous convolution (DMAC)	Robust under geometrical attacks	High computational time, difficult to spot regions under transformations
Abdlla [155]	MNIST, CIFAR-10	CMF	VGG-16 network	CNN and Generative adversarial network (GAN)	High detection accuracy and other forgery types	Need to test on more images
Jabeen [156]	MFC NIST 2018, CASIA v2.0, CoMo-FoD	CMF and Image splicing	Inception-V3	Deep neural networks (DNN)	First method to combine forgery detection systems with provenance filtering	Accuracy needs to be enhanced
Elaskily [157]	MICC-F220, MICC-F600, MICC-F2000, SATs-130	CMF	Convolutional layers followed by max-pooling	CNN	Good performance parameters	Low speed and need to test on more datasets
Ahmed [158]	Computer-generated using COCO dataset	Image splicing	Feature pyramid network	Mask-RCNN	Faster convergence	Provide less accuracy and need to generalize concept to other forgery types

		Actual	
		Positive	Negative
Predicted	Positive	True Positive (T_{PO})	False Positive (F_{PO})
	Negative	False Negative (F_{NE})	True Negative (T_{NE})

Fig. 18 Confusion matrix

total images recognized incorrectly as tampered, F_{NE} depicts total tampered pictures that were missed, and T_{NE} signifies total authentic pictures impeccably recognized as genuine.

Detection accuracy is sum of T_{PO} and T_{NE} to entire pictures. Precision is a term used to describe the likelihood that the identified picture is genuinely forged. Specificity or TNR (true negative rate) is the measure of an image's ability to be recognized as original. False positive rate is forged pictures that cannot be detected (FPR). The correlation between actual and anticipated classes is measured by Mathew's correlation coefficient. The likelihood that classifier is informed about the condition is stated as informedness, while the probability that the circumstance is marked by classifier is stated as markedness [1, 126, 159, 160]. Below are the formulae for some widely used performance measures.

$$DA = \frac{T_{PO} + T_{NE}}{F_{PO} + T_{PO} + F_{NE} + T_{NE}} \quad (1)$$

$$P = \frac{T_{PO}}{T_{PO} + F_{PO}} \quad (2)$$

$$TPR = R = Sensitivity = \frac{T_{PO}}{T_{PO} + F_{NE}} \quad (3)$$

$$TNR = Specificity = \frac{T_{NE}}{T_{NE} + F_{PO}} \quad (4)$$

$$F_1 = 2 \frac{P \cdot R}{P + R} \quad (5)$$

$$FNR = \frac{F_{NE}}{T_{PO} + F_{NE}} \quad (6)$$

$$FPR = \frac{F_{PO}}{T_{NE} + F_{PO}} \quad (7)$$

$$F_2 = 5 \frac{P \cdot R}{4 \cdot P + R} \quad (8)$$

$$\text{Informedness} = \text{TPR} + \text{TNR} - 1 \quad (9)$$

$$\text{Markedness} = \frac{T_{PO}}{T_{PO} + F_{PO}} + \frac{T_{NE}}{T_{NE} + F_{NE}} - 1 \quad (10)$$

$$\text{MCC} = \frac{T_{PO} \times T_{NE} - F_{PO} \times F_{NE}}{\sqrt{((T_{PO} + F_{PO})(T_{PO} + F_{NE})(T_{NE} + F_{PO})(T_{NE} + F_{NE}))}} \quad (11)$$

7 Challenges and Future Directions

Following a thorough evaluation of the literature gathered systematically, it has been discovered that still considerable work needs to be accomplished in this sector, as well as several issues must be addressed. To summarise, prevalent obstacles include computational complexity and feature dimensionality, robustness against extensive post-processing procedures and geometrical transformation invariance, pinpointing the precise location of the forgery, the absence of a dataset encompassing all potential forging attempts, and the reliance on human interpretation. Thus, numerous future directions for this study are presented in this section as below, based on the existing challenges in the current research state.

- **Benchmark dataset:** The choice of datasets is an important consideration in assessing the proposed method. Datasets like Image manipulation already address pre-processing procedures like JPEG compression, rotation, noise addition, and scaling. Yet, there is still room for development. To date, processes such as quantization at various stages, blurring, Gaussian and white noise, and other similar operations have been conducted through the pre-processing stage. However, it results in distinct datasets for numerous authors, resulting in a loss of consistency. In this scenario, the experiments cannot be efficiently repeated. As a result, the claims of accuracy made by different academics are incomparable because they utilized different compression and scaling techniques. As a result, in the future, a dataset having all of these inherent attributes must be presented.
- **Feature dimensionality:** In the present research papers, feature dimensionality and computational complexity are the most common challenges. So, in the future, to minimize feature dimensionality, efficient feature reduction procedures that can keep feature qualities even when reduced to a given dimension must be created. Thus, the reduction of features may result in reduced computational complexity.
- **Deep learning:** Other advances in the feature extraction phase are predicted. There are just a few deep learning-based IFDT. Deep learning is only employed in the categorization of genuine and counterfeit pictures. Because of the differences in the training and testing sets, as well as the complexity of the experiments, these IFDT based on deep learning are hard to repeat and compare. In the future, the researchers will investigate this subject utilizing deep learning technologies like CNN. Since CNN and transfer learning-based schemes are predicted to outperform the handcrafted features-based models.

- **Robustness:** Numerous established methodologies exhibit deficiencies in their ability to withstand transformations such as blurring, scaling, noise introduction, JPEG compression, and rotation. Thus, to be resistant to numerous post-processing processes and geometrical changes, statistical characteristics must be resilient.
- **Reinforcement learning:** Another predicted advancement in the future is reinforcement learning, which will replace the requirement for human interaction in classifications. The machine learns through its interactions with the environment in reinforcement learning. It receives both good and negative responses from its surroundings and therefore acquires throughout period. It behaves similarly that an analyst investigates and acquires through experience.
- **Localization:** Furthermore, only a few of the existing research studies for IFDT can locate the tampered areas. As a result, future studies could focus on developing a tool that can automatically detect the type of forgery employed in an image. To increase the localization of fabricated areas, a multi-scale analysis may be expanded to multiple forensic methodologies.

Several scientific advancements in dataset creation, feature extraction, and feature classification are predicted soon. As a result, it is believed that digital image forensics is an intriguing topic of research with many outstanding challenges that will undoubtedly witness intense research activity in the near future.

8 Conclusion

Digital image passive forensics technology is one of the fastest expanding topics of study. Nowadays, passive image forgery detection procedures are still in the early stages of development, according to this evaluation, and have a huge amount of potential for future research and development applications. Therefore, this study focuses on examining passive techniques to image forgery detection. The categorization of IFDT is discussed in this study using a suitable taxonomy. It also describes the survey's review method, which includes research inquiries and information sources. Existing benchmark datasets are also examined for relevance. A thorough examination and critique of passive techniques are also provided. The performance assessment criteria for assessing the performance of passive systems are provided. In addition, directions for future research are also presented. It is expected that this survey would give important information to academics, scientists, and related research societies on this subject. The study in image forensics is currently ongoing and will remain to investigate forensics methods with great accurateness and consistency.

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Declarations

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